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Department of Industrial Economics and Technology Management

Examination paper for TIØ4145 Corporate Finance

Examination date: 18.12.2023

Examination time (from-to): 15.00-19.00

Permitted examination support material: D / No printed or hand-written support material is allowed. A specific basic calculator is allowed. Formula sheet is provided in Inspera.

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OTHER INFORMATION

Get an overview of the question set before you start answering the questions.
You will find a formula sheet down to the left corner.

Read the questions carefully and make your own assumptions. If a question is unclear/vague, make your own assumptions and specify them in your answer. The academic person is only contacted in case of errors or insufficiencies in the question set. Address an invigilator if you suspect errors or insufficiencies. Write down the question in advance.

Hand drawings: For questions in Sections 2, 3 and 4, you may choose to answer on handwritten sheets. At the bottom of the question you will find a seven-digit code. Fill in this code in the top left corner of the sheets you wish to submit. We recommend that you do this during the exam. If you require access to the codes after the examination time ends, click “Show submission”.

Weighting: Weighting is given in the question set. Section 1 consists of multiple choice questions. The multiple choice questions are graded as follows: 2 points are given for correct answer; 1 point is subtracted for wrong answer; 0 points is given when the question is left unanswered. Only one answer is correct.

Notifications: If there is a need to send a message to the candidates during the exam (e.g. if there is an error in the question set), this will be done by sending a notification in Inspera. A dialogue box will appear. You can re-read the notification by clicking the bell icon in the top right-hand corner of the screen.

Withdrawing from the exam: If you become ill or wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click “Submit blank”. This cannot be undone, even if the test is still open.

Access to your answers: After the exam, you can find your answers in the archive in Inspera. Be aware that it may take a working day until any hand-written material is available in the archive.

1 Financial Options

Which of the following statements about option pricing is FALSE?

Select one alternative:

- The option delta, Δ, has a natural interpretation: It is the change in the price of the stock given a $1 change in the price of the option.
- Because a leveraged position in a stock is riskier than the stock itself, this implies that call options on a positive beta stock are more risky than the underlying stock and therefore have higher returns and higher betas.
- Only one parameter input for the Black-Scholes formula, the volatility of the stock price, is not observable directly.
- We do not need to know the expected return on the stock to calculate the option price in the Black-Scholes Option Pricing Model.

Maximum marks: 2

2 The Flow-to-Equity Method

Which of the following is NOT a step in the valuation process using the flow to equity method?
Select one alternative:

- Compute the equity value, E, by discounting the free cash flow to equity using the equity cost of capital.
- Determine the equity cost of capital, r_E.
- Determine the before-tax cost of capital, r_U.
- Determine the free cash flow to equity of the investment.

Maximum marks: 2

3 Project Risk Characteristics and Financing

In a world with taxes, which of the following is the rate we should use to evaluate a project with the same risk and the same financing as the firm itself?
Select one alternative:

- The cost of equity
- The weighted-average cost of capital
- The pre-tax WACC
- The cost of debt

Maximum marks: 2

4 Beta

Which of the following statements about the beta is FALSE?
Select one alternative:

- One difficulty when trying to estimate beta for a security is that beta depends on the correlation and volatilities of the future returns of the security and the market.
- Securities that tend to move less than the market have betas below 1.
- It is common practice to estimate beta based on the expectations of future correlations and volatilities.
- Beta is a measure of systematic risk

Maximum marks: 2

5 Dynamic Behavior of Bond Prices

Which of the following statements about bonds is TRUE?
Select one alternative:

- If a bond trades at a discount, its yield to maturity will exceed its coupon rate.
- When a bond is trading at a discount, the price drop when a coupon is paid will be larger than the price increase between coupons.
- As interest rates and bond yields fall, bond prices will fall.
- Bonds with higher coupon rates are more sensitive to interest rate changes.

Maximum marks: 2

6 Sovereign bonds

A key difference between sovereign default and corporate default is:
Select one alternative:

- Unlike corporate debt, sovereign debt prices are not inverse to yields.
- Unlike a corporation, a country facing difficulty meeting its financial obligations typically has the option to print more currency.
- Unlike a corporation, a country facing difficulty meeting its financial obligations cannot default.
- Unlike a corporation, any country can turn to the European Monetary Union to pay off its debts.

Maximum marks: 2

7 Discounted Free Cash Flow Model

Which of the following statements about the Discounted Free Cash Flow model is FALSE?
Select one alternative:

- When using the discounted free cash flow model, we forecast the firm's free cash flow up to some horizon, together with some terminal (continuation) value of the enterprise.
- We estimate a firm's current enterprise value by computing the present value of the firm's free cash flow.
- Because the firm's free cash flow is equal to the sum of the free cash flows from the firm's current and future investments, we can interpret the firm's enterprise value as the total NPV that the firm will earn from continuing its existing projects and initiating new ones.
- Free cash flow measures the cash generated by the firm after payments to debt or equity holders are considered.

Maximum marks: 2

8 CAPM

In the capital asset pricing model, if the expected market return is below the risk-free rate, what implication does this have for the security market line (SML)?

Select one alternative:

- SML becomes irrelevant
- SML will have a positive slope
- SML will have a negative slope
- SML will be horizontal

Maximum marks: 2

9 Optimal Capital Structure with Market Imperfections

Which of the following statements is FALSE?

Select one alternative:

- In general, as a firm's interest expense approaches its expected taxable earnings, the marginal tax advantage of debt increases.
- Increasing the level of debt increases the probability of bankruptcy.
- With perfect capital markets, the risk of bankruptcy is not a disadvantage of debt—bankruptcy simply shifts the ownership of the firm from equity holders to debt holders without changing the total value available to all investors.
- Aside from taxes, another important difference between debt and equity financing is that debt payments must be made to avoid bankruptcy, whereas firms have no similar obligation to pay dividends or realize capital gains.

Maximum marks: 2

10 Efficient portfolio

Which of the following statements about the efficient portfolio is FALSE?

Select one alternative:

- Because every security is owned by someone, the sum of all investors' portfolios must equal the portfolio of all risky securities available in the market.
- According to CAPM, the efficient portfolio, the portfolio that all investors should hold, must be the same portfolio as the market portfolio of all risky securities.
- If all investors demand the efficient portfolio, and since the supply of securities is the market portfolio, then two portfolios must coincide.
- If some security were not part of the efficient portfolio, then every investor would want to own it, and demand for this security would increase causing its expected return to fall until it is no longer an attractive investment.

Maximum marks: 2

11 Question 2.1 (10%)

Following data are given:

	Expected return	Variance of Returns	Covariance between stock returns and market returns
Stock 1	0.13	0.01	0.032
Stock 2	0.22	0.09	0.036
Market	0.18	0.04	0.040
Risk Free Rate	0.06	0	0

i) Based on this data, write an equation of a Capital Market Line (CML) that follows from the CAPM and draw a diagram of CML.

ii) Show the position of the two stocks, market portfolio and the risk free rate relative to the Capital Market Line in your drawing.

iii) Comment on the position of the two stocks relative to the CML and explain what it implies.

Fill in your answer here

Format

Words: 0

Maximum marks: 10

12 Question 2.2 (10%)

i) Based on the data from the table in Question 2,1, write an equation of a Security Market Line (SML) that follows from the CAPM and draw a diagram of SML.

ii) Show the position of the two stocks, market portfolio and the risk free rate relative to the SML in your drawing

iii) Comment on the position of the two stocks relative to the CML and explain what it implies.

Fill in your answer here

Format

Words: 0

Maximum marks: 10

13 Question 2.3 (5%)

i) Is market portfolio always efficient in reality? Explain why.

ii) Name one challenge associated with testing the efficiency of the market portfolio in real-world scenarios.

Fill in your answer here

Format

Words: 0

Maximum marks: 5

14 Question 3.1 (10%)

The following table summarizes prices of various default-free zero-coupon bonds (expressed as a percentage of face value):

Maturity (years)	**1**	**2**	**3**	**4**	**5**
Price (per $100 face value)	94.52	89.85	85.89	82.27	79.10

i) Find yield to maturity for each of these five bonds. For this question, please round your answers to two decimals.

ii) What can you say about the slope of the zero-coupon yield curve? What does the shape of this yield curve indicate about the relationship between the bond maturities and their yields? Discuss the potential economic implications of such a yield curve.

Fill in your answer here

Format

Words: 0

Maximum marks: 10

15 Question 3.2 (10%)

Use the data from Question 3.1.

i) Find the price today of a 3-year default-free security with a face value of $1000 and an annual coupon rate of 6%.

ii) Does this security trade at a premium, at a discount or at par?

iii) Illustrate in a graph the effect of the passage of time on bond price for this bond assuming that the yield remains constant.

Fill in your answer here

Format

Words: 0

Maximum marks: 10

16 Question 4.1 (20%)

Fabulosa Pizza is active in the fast food business. It currently has a book value of $80 million, with equal proportions on debt and equity. The firm has 10 million shares outstanding, which currently sell for $6 per share. The outstanding debt of Fabulosa is frequently renegotiated and adapted to the market conditions, so that capital structure remains constant. The firm has a tax rate of 40%. The risk-free rate is equal to 6%. The return on the market portfolio is 14%. Fabulosa wants to expand its business into offshore salmon farming. This will require an investment of $750 million and is expected to produce a perpetual yearly cashflow of $100 million. The company wants to finance the investment with 25% debt, for which the bank made an offer at 8% interest. At present, there is only one firm active in offshore salmon farming business, OffSea AS. OffSea As is financed with equal parts of debt and equity, its debt has a 9% interest rate and its equity has a beta of 1.4. Fabulosa will keep debt at 25% of the project value.

i) Should Fabulosa Pizza accept the project or not? Use calculations to support your answer.

ii) What effect will the decision under a) have on Fabulosa's stock price? Explain why (no calculations are required)

Fill in your answer here

Format

Words: 0

Maximum marks: 20

17 Question 4.2 (10%)

Assume now that Fabulosa owners care about environment and want to incorporate environmental value in their analysis. They estimated that the offshore salmon farming facility will generate carbon emissions that amount to 30 000 tons of CO2 equivalent per year in perpetuity. Fabulosa assumes shadow carbon price of $200 per 1 ton of CO2 equivalent.

i) Make your own reasonable assumption regarding the discount rate Fabulosa should use when calculating environmental value. Comment whether it should be the same, smaller or larger than the discount rate used in Question 4.1.

ii) Should Fabulosa undertake the project then?

Fill in your answer here

Format

Words: 0

Maximum marks: 10

18 Question 4.3 (5%)

Name two reasons why sustainability considerations are traditionally excluded from financial analysis.

Fill in your answer here

Format

Words: 0

Maximum marks: 5